

YUBIN KIM

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RESEARCH INTERESTS

- Agentic AI, Multi-Agent Systems, Efficient AI, Sequence Modeling, Machine Learning, Deep Learning, Time Series Modeling, Efficient Transformers

EDUCATION

Massachusetts Institute of Technology

S.M./Ph.D. in Operations Research

- Advisor: Prof. Giannis Daras

Cambridge, MA

Sep 2026 – May 2031 (Projected)

Georgia Institute of Technology

B.S. Industrial Engineering, GPA: 4.0/4.0, Highest Honor

- Concentration: Advanced Studies in Operations Research & Statistics (PhD-track)
- Selected Coursework: Machine Learning, Probability & Statistics, Optimization, Stochastic Processes, Regression & Forecasting
- Honors: ISyE Alpha Pi Mu Academic Excellence Award, Pennington Undergraduate Research Award (PURA), Dean's List (All Semesters)
- Global Leadership Living Learning Community

Atlanta, GA

Aug 2023 – Dec 2025

RESEARCH EXPERIENCE

AI for Time Series Lab, Georgia Tech

Undergraduate Researcher (Advisor: Prof. Shihao Yang; IE-PURA Recipient)

- Developing novel architectures for multivariate time series forecasting using efficient Transformer-based models in PyTorch.
- Co-authored a NeurIPS 2025 Spotlight paper (*ZeroS: Zero-Sum Linear Attention for Efficient Transformers.*) proposing a zero-sum linear attention mechanism for efficient long-sequence modeling.
- First-authored *StretchTime: Adaptive Time Series Forecasting via Symplectic Attention* (ICML 2026), introducing a trainable temporal embedding mechanism.
- Co-authored *CAPS: Unifying Attention, Recurrence, and Alignment in Transformer-based Time Series Forecasting* (submitted to NeurIPS 2026; available on arXiv).
- Led an interdisciplinary research team through comprehensive literature review, novel architecture conceptualization, and iterative refinement via rigorous ablation and scaling experiments.

Atlanta, GA

Aug 2024 – Present

Financial Engineering Lab, Yonsei University

Research Intern (Advisor: Prof. Kyoungju Oh)

- Developed deep learning models (LSTM, CNN, Transformer) for market prediction and volatility forecasting.
- Processed large-scale, high-frequency market data using Pandas and NumPy to construct predictive financial features.
- Evaluated model performance under different market regimes and feature sets; presented results to faculty and researchers.

Seoul, KR

May 2024 – Jul 2024

PUBLICATIONS

- J. Lu, X. Han, Y. Sun, V. Pati, **Y. Kim**, S. Somani, S. Yang. *ZeroS: Zero-Sum Linear Attention for Efficient Transformers*. In *Advances in Neural Information Processing Systems* (NeurIPS 2025), Spotlight.
- **Y. Kim**, V. Pati, J. Twitty, V. Pham, S. Yang, J. Lu. *StretchTime: Adaptive Time Series Forecasting via Symplectic Attention*. In *Proceedings of the International Conference on Machine Learning* (ICML 2026).
- V. Pati, **Y. Kim**, V. Pham, J. Twitty, S. Yang, J. Lu. *CAPS: Unifying Attention, Recurrence, and Alignment in Transformer-based Time Series Forecasting*. Preprint (Under review for NeurIPS), 2026.
- Y. Kim, **Y. Kim**. *미국 주식으로 시작하는 슬기로운 퀀트투자* [Smart quantitative investing in U.S. stocks: A beginner-friendly Python guide] Hanbit Media, 2021.

TEACHING EXPERIENCE

ISYE 3133: Engineering Optimization, Georgia Tech

Atlanta, GA

Undergraduate Teaching Assistant

Jan 2025 – May 2025

- Assisted instruction for 150+ students in linear, discrete, and combinatorial optimization.
- Conducted weekly recitations and office hours on simplex method, branch-and-bound, and duality.
- Created grading rubrics and problem sets in LaTeX and Python to assess conceptual mastery.

SELECTED PROJECTS

Sepsis Care Optimization with Emory Healthcare

Atlanta, GA

Team Lead, Georgia Tech Senior Design (ISYE 4106)

Aug 2025 – Present

- Leading a collaboration with Emory Healthcare's Sepsis Care Improvement Program to improve SEP-1 compliance through machine learning and workflow optimization.
- Developing a real-time patient condition scoring system using XGBoost, LSTM, and GRU models trained on irregularly updated EHR data.
- Implementing a survival analysis pipeline to project patient risk trajectories under sparse temporal sampling.
- Optimizing resource allocation by identifying the ultrasound machine bottleneck through workflow simulation and operations research models.
- Expected to directly contribute to Emory's compliance improvement efforts, preventing up to 2% reimbursement losses from CMS noncompliance.

Evolution of NLP Models for Authorship Classification

Atlanta, GA

Team Lead, Methods & Applications of Machine Learning (ISYE 4600)

Jan 2025 – May 2025

- Investigated the evolution of NLP techniques through the task of human vs. AI text discrimination.
- Benchmarked traditional ML models (Random Forest, Logistic Regression) against deep models (FastText + RNN) and modern embeddings (BERT + CatBoost).
- Demonstrated the superiority of contextual embeddings in distinguishing GPT-generated content, achieving strong interpretability of linguistic shifts.
- Strengthened theoretical and practical understanding of representation learning and transfer learning in NLP.

LEADERSHIP & COMMUNITY SERVICE

- **Chief Operating Officer, UMarket** — Managed a 13-member cross-functional team to build an online marketplace for university communities; led strategy and fundraising.
- **Consultant, Epic Intentions** — Delivered strategic recommendations for non-profits in Ghana and the U.S. through data-driven business analysis.
- **Global Leadership Living Learning Community** — Created a NIH R21-style grant proposing a solution the lack of digital literacy in sub-Saharan Africa, contributing to the UN Sustainable Development Goals.

TECHNICAL SKILLS & COURSEWORK

- **Programming:** Python (PyTorch, TensorFlow, scikit-learn, NumPy, Pandas), R, Java
- **Tools:** Git, LaTeX, Linux, Jupyter, Gurobi
- **Topics:** Sequence Modeling, Machine Learning, Time Series, Optimization, Stochastic Processes, Statistics

HONORS & AWARDS

- NeurIPS 2025 Spotlight
- ISyE Alpha Pi Mu Academic Excellence Award
- ISyE Pennington Undergraduate Research Award (IE-PURA), Georgia Tech, Spring 2025
- Dean's List (All Semesters)